Fundamentals in AI and ML

PROJECT REPORT

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1. Project Title

House Price Predictor using Machine Learning (Linear Regression)

2. Problem Overview

The decisions made by people regarding real estate prices are usually based on estimations or assumptions. This can sometimes result in incorrect financial decisions for people, especially small real estate dealers. The problem statement for the project is to predict the house price using machine learning techniques.

The project will predict the expected price for the house/flat based on the following parameters:

Area (carpet area in square feet)

Number of BHKs

Number of bathrooms

Resale or not

Location

Floor information

Furnished or not

3. Why This Problem Matters

The problem is significant from the real-world perspective. The predictions made by the model will be helpful for people who want to invest in real estate. The predictions will assist people in making correct financial decisions. The predictions will not only be helpful for people who want to invest in real estate, but they will also be helpful for people who want to sell their properties.

The project is significant from the academic perspective. The project will serve as an example for students who are learning about machine learning. The project is significant from the academic perspective because it is an example for supervised learning.

4. Project Objectives

The objectives of the project are as follows:

Develop a machine learning model using the regression algorithm to predict house prices.

The project will consider both numerical and categorical values.

The model will be saved, and the predictions will be made using the command-line interface.

The project will be developed in such a manner that it is easy to understand for students.

5. Scope and Boundaries

The scope is as follows:

Offline training using the prepared CSV file.

- Feature Preprocessing and Model Training

- Price Prediction Using User Input from Command Line

Out of scope:

- Web/mobile UI

- Live market feed and updates

- Legal/financial advice

- Hyper-advanced ensemble methods and deep learning

6. Dataset and Preparation

The workflow described in this project uses a Kaggle housing dataset. A cleaned version of this dataset is used as a dataset/dataset.csv for training.

Columns used for training:

- Target column: Price

- Numeric features: BHK, Area, Is_Resale, Bathrooms

- Categorical features: Location, Floor, Furnishing

Preparation strategy used in this project, according to the project notes, includes:

- Deleting rows with invalid values

- Converting price strings to numeric

- Useful values extracted: BHK, area

- Only using relevant columns for training

7. Approach and Methodology

The project uses a Supervised Regression approach. The implementation is in the file train.py.

7.1 Model Design

- Algorithm: Linear Regression

- Reason: Simple algorithm with easy-to-understand behavior, low training complexity, and suitable for academic purposes

7.2 Preprocessing Strategy

- Numerical values passed through without scaling

- Categorical values transformed using OneHotEncoder

- handle_unknown='ignore' enabled to avoid crashes

7.3 Pipeline Integration

- A scikit-learn Pipeline is used to integrate regression and preprocessing

7.4 Model Persistence

- After training, the pipeline is serialized with pickle and saved to model/model.pkl

8. Implementation Summary

8.1 Training Script (train.py)

- Loads dataset/dataset.csv

- Splits input features and target column Price

- Creates ColumnTransformer to handle mixed feature types

- Trains LinearRegression in a pipeline

- Saves the trained model to disk as model/model.pkl

8.2 Prediction Script (predictor.py)

- Loads the saved model from disk as model/model.pkl

- Asks users to input data via command line

- Converts necessary numerical inputs to float type

- Converts the input of Resale into binary form

- Creates a data frame with the expected format

- Returns the price prediction in the expected format of rupees

9. Key Decisions and Rationale

1. Algorithm Used: Linear Regression

Reason: A good starting point and easy to explain in a student project.

2. Use of OneHotEncoder on Category Columns

Reason: Needed to encode the text data in Location, Floor, and Furnishing columns into a numeric array to be consumed by Linear Regression.

3. Use of a Scikit-Learn Pipeline

Reason: To ensure train and predict preprocessing steps are consistent and to make the code more portable across different versions of scikit-learn.

4. Command Line Interface Approach

Reason: A faster implementation to focus on the core logic of the code first, and then move to the GUI later if needed.

5. Use of Pickle to Serialize and Deserialize the Model

Reason: The most straightforward and commonly accepted method of saving and loading scikit-learn models.

10. Challenges Faced

1. Complexity of Data Cleaning

Raw data contains mixed data types and missing values, and the text data in Location and other fields have different patterns. This had to be standardized to prevent errors in training the model.

2. Complexity of Dealing with Categorical Diversity

Location and other fields have many possible values. This had to be taken into account to make the model's prediction reliable.

3. Input Data Reliability in Command Line Interface

Since the prediction relies on the inputs provided by the users, some basic validation is included. This is a weak point and could be strengthened further to make the code more robust.

4. Trade-Off between Model Complexity and Accuracy

Linear Regression is simple but might not capture the true complexity of the relationships in the data, which might be nonlinear in the case of real estate. This is a commonly accepted trade-off in this version of the code.

11. Results and Observations

- The project is successfully training and saving a reusable model.

- The predictor script is successfully generating an estimated property price based on user input details.

- The project is successfully showing end-to-end functionality from training to live command-line inference.

General observations:

- Predictions are dependent on the quality of the input data provided and the representation of the input features.

- Quality of categorical features, especially the level of granularity for the location, significantly affects the usability of the predicted results.

- The model is successfully performing its purpose as an initial or base estimation tool, not necessarily an authority tool.

12. Limitations

- There is no formal training, testing, or evaluation metrics printed out within the current predictor script

- There is no cross-validation currently implemented within the project

- There is limited error checking for incorrect or non-numeric values within the input data provided to the model

- There is no feature engineering included for advanced input features (e.g., age of property, direction it faces, amenities, proximity, etc.)

- The dynamic nature of market conditions means that the training data will become outdated over time

13. What I Learned

This project has allowed me to apply learning to the entire machine learning process:

- The process of converting real-world problems into supervised learning problems

- The importance of data cleaning prior to model training

- Combining data preprocessing with model training using ColumnTransformer and Pipeline

- The importance of saving models for future reuse

- The need for user input interface schema to perfectly align with the model's expected input features

- The difference between having a working model versus having a working model that is deployable

14. Future Improvements

- Include training, testing, and evaluation metrics (MAE, RMSE, R2)

- Compare different regressors (Ridge, Lasso, Random Forest, XGBoost, etc.)
- Improve input validation and exception handling within predictor.py
- Introduce feature engineering and handling of outliers
- Design a lightweight web interface using Flask or Streamlit for wider usability
- Include tests for the training and prediction scripts

15. Conclusion

The House Price Predictor project has effectively demonstrated the complete machine learning baseline solution for property price prediction using mixed feature data in tabular format. The project is applicable in the real world and represents a complete process from data preparation to model training and interactive prediction.

The system, although in its current simple form, represents a good foundation for further development in model evaluation and robustness. The project has achieved the primary goal of creating a clear, functional, and educational end-to-end machine learning solution.